

Week 4: ETL in Azure Databricks

Objective

Perform ETL (Extract, Transform, Load) for an Expense Monitoring System using Azure Databricks.

Capstone Tasks

- Upload cleaned data into Databricks
- Combine user and expense data
- Create a summary table with monthly spend, savings, and alerts
- Save the final report as Delta or CSV for dashboards

Tools Used

- Azure Databricks
- PySpark

Input Files

users.csv

user_id,name,income

1,Thamarai,10000

2,Ravi,12000

3,Priya,8000

expenses.csv

user_id,date,amount

1,2026-05-01,500

1,2026-05-15,5000

2,2026-05-03,400

2,2026-05-20,4500

3,2026-05-07,200

3,2026-05-18,250

ETL Steps

Step 1: Upload Data to Databricks

Upload users.csv and expenses.csv to Databricks FileStore (example path: /FileStore/tables/)

Step 2: Load Data Using PySpark

```
from pyspark.sql.functions import sum, col, when, month

users = spark.read.csv("/FileStore/tables/users.csv", header=True, inferSchema=True)

expenses = spark.read.csv("/FileStore/tables/expenses.csv", header=True,
inferSchema=True)

users.show()

expenses.show()
```

Step 3: Transform Data

Add month column and calculate monthly spending.

```
expenses = expenses.withColumn("month", month(col("date")))

monthly_spend = expenses.groupBy("user_id", "month") \
.agg(sum("amount").alias("monthly_spend"))
```

Step 4: Combine User and Expense Data

```
summary = users.join(monthly_spend, "user_id")
```

Step 5: Calculate Savings and Alerts

```
summary = summary.withColumn(
"savings",
col("income") - col("monthly_spend")
)
```

```
summary = summary.withColumn(  
  "alert",  
  when(col("monthly_spend") > col("income") * 0.7, "High Spending")  
    .otherwise("Normal")  
)  
summary.show()
```

Step 6: Save Final Report

Save as Delta:

```
summary.write.format("delta").mode("overwrite").save("/FileStore/final_report_delta")
```